

# Genetic Programming Hyper-Heuristics for the Job Shop Scheduling Problem with Interval Durations: Robustness-Aware Interpretable Rules that Generalize Across Instance Sizes

Hernán Díaz Rodríguez

*Department of Computing, University of Oviedo, Campus de Gijón, Gijón, 33204, Spain*

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## Abstract

In the job shop scheduling problem with interval durations (IJSP), processing times are uncertain and represented by an interval. We implement a genetic programming hyper-heuristic to evolve dispatching rules for this problem, which build a complete schedule in a single pass. Each rule is a short, readable formula that ranks the candidate operations using both the upper bounds and the widths of the intervals. On the 70 interval Taillard instances, the 30 rules from independent evolutions average 18.99% ( $\pm 1.33$ ) relative error over reference bounds. The rule selected on the validation set attains 17.71% and improves on the strongest constructive baseline, Giffler–Thompson/MWKR (29.5%), on every instance. Evolved on four  $20 \times 15$  instances, the rules transfer zero-shot to all size classes and to 12 classical instances. Against a published genetic algorithm reimplemented on the same decoder, the rule sampled repeatedly attains lower error up to a budget that grows with instance size, from 10 to 300 s per instance. Seeding the genetic algorithm with the rule also speeds up its early search.

Ablations delimit what interval awareness contributes. Under the default expected-makespan objective, rules evolved without the width terminals or on crisp midpoints do equally well; its value lies in robustness: an objective penalizing interval width yields narrower predicted intervals and executions that deviate less from the prediction, a gain that requires the width terminals and holds on asymmetric intervals, at the cost of a higher expected makespan.

*Keywords:* interval job shop scheduling, genetic programming, hyper-heuristics, dispatching rules, scheduling under uncertainty, robustness

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## 1. Introduction

In real manufacturing environments, the exact duration of an operation is rarely known in advance [1]. The interval job shop scheduling problem (IJSP) models this uncertainty without assuming any probability distribution: each processing time is given by an interval with known lower and upper bounds. Schedules are built with interval arithmetic, so the makespan of a solution is itself an interval, and solutions are compared with an interval ranking. Metaheuristics for the IJSP, from genetic algorithms to artificial bee colony algorithms hybridized with local search, achieve strong results by building and evaluating large numbers of complete solutions for every instance [2–5].

At the opposite end of the computational spectrum are *dispatching rules*: priority functions that build a schedule in a single constructive pass by repeatedly selecting the next operation to start. They combine speed, interpretability and ease of deployment, and their quality rests on the priority criterion, which is hard to design by hand.

For *deterministic* scheduling, genetic programming (GP) designs these criteria automatically, and the evolved rules have been shown to outperform hand-crafted ones [6, 7]. For scheduling under interval uncertainty, however, the automated design of dispatching rules remains, to the best of our knowledge, unexplored. Moving automated rule design to interval durations raises three questions that do not arise in the deterministic case. First, which attributes does the evolution select when the duration, earliest start and remaining work of an operation are intervals rather than numbers? Second, is the width of the intervals useful for dispatching, or does their position alone drive the decision? Third, does an objective that rewards a narrow makespan interval lead to schedules whose execution departs less from the prediction? Answering them requires exposing the uncertainty to the rule as attributes of its own, which is the starting point of this work.

A fourth question is where such rules stand among the methods already available for the IJSP. Rules and metaheuristics are usually compared at their natural operating points, one constructive pass against a complete search. That comparison ranks them, but does not say for which computational budget each is preferable, so we compare them on a common budget axis

within a single implementation. We do not reopen how interval makespans should be compared, since they admit no natural total order: we adopt the ranking that a systematic study of the alternatives found most robust [2].

This paper makes the following contributions:

1. A GP hyper-heuristic that brings automated rule design to the IJSP: priority expressions over a terminal set in which the duration, earliest start and work remaining of a candidate operation appear both as worst-case values and as widths, with fitness computed by interval arithmetic. The rules it produces are compact enough to be simplified and read by hand (Sections 4 and 7.1).
2. An empirical study of the evolved rules on the 70 interval Taillard instances comprising (i) a comparison with deterministic constructive methods at equal computational cost and (ii) an evaluation of zero-shot generalization across instance sizes and to a second benchmark of classical instances (Section 6).
3. A comparison with per-instance search: against the published results of the IJSP metaheuristics, and at matched budget against our Python implementation of the published genetic algorithm, with and without the rule as a seed. The rule, its sampled variant GP-best-of- $N$  and the genetic algorithm share decoder and evaluator, and are compared across budgets in seconds per instance on instances of different sizes (Sections 6.4 and 6.5).
4. A delimitation of what interval awareness contributes. Under the expected-makespan objective, ablations remove the width terminals or evolve on crisp midpoint instances (Section 7.4). An objective that penalizes interval width, which has no crisp counterpart, is then examined for its effect on the deviation of the executed makespan from its prediction, as a function of the penalty weight, under several realization laws and on asymmetric intervals (Sections 7.5–7.7).

Section 2 reviews related work and Section 3 formalizes the IJSP. Section 4 presents the GP hyper-heuristic and Section 5 the experimental setup. Section 6 reports the results, and Section 7 analyzes them and states the limitations. Section 8 concludes.

## 2. Related Work

### 2.1. Job shop scheduling under interval uncertainty

Interval durations offer a distribution-free uncertainty model that requires only bounds, in contrast to stochastic and fuzzy approaches [8]. They are the natural choice when historical data are insufficient to estimate distributions and a decision maker can only supply a minimum and a maximum duration. Scheduling with interval processing times has been studied from the single machine [9] to the flexible job shop with interval or grey processing times [10, 11], and Allahverdi [1] surveys the flow shop and other environments. Recent work extends the model to assembly [12] and energy-aware flexible [13] shops. Sotskov et al. [14] study the problem analytically, through measures of the uncertainty induced by the intervals and of the stability of optimal sequences. Sotskov et al. [15] examine the execution of schedules built under uncertain durations, the question Section 7.6 examines for evolved rules. Calamita et al. [16] quantify the risk of a fixed schedule directly: they estimate the value-at-risk and conditional value-at-risk of its makespan when only the range of each duration is known. Machine-learning predictors built on features of the activity network replace the costly exact evaluation.

For the job shop with interval durations and makespan minimization, which concerns us here, the literature is dominated by population-based meta-heuristics. Lei’s population-based neighborhood search [17], which ranks interval makespans by a possibility degree, was an early reference point. It was later outperformed by a genetic algorithm with a systematic study of interval rankings [2] and by artificial bee colony variants [3–5], which report the best published results for this problem. Related objectives have also been addressed under interval uncertainty, notably tardiness minimization [18]. All of these methods build, evaluate and refine many complete solutions for each instance. Fast constructive methods for the IJSP have received far less attention: dispatching rules adapted to intervals serve mainly as baselines, which is also their role in this paper (Section 6.2). Constructive heuristics are also a standard source of initial solutions for population methods [19]. Rules evolved by GP have been used in this way to initialize a cooperative evolutionary algorithm for a deterministic flexible job shop [20], a use Section 6.5 examines under interval uncertainty.

## 2.2. Dispatching rules for job shop scheduling

Priority dispatching rules are among the oldest and most widely deployed scheduling heuristics. Classical rules score each ready operation by a single attribute, and we adopt their standard acronyms throughout this paper. *Shortest* and *longest processing time* (SPT, LPT) look at the duration of the operation itself. *Earliest starting time* (EST) prefers the operation that can begin soonest given the current state of its job and its machine. *Most work remaining* (MWKR) and *most operations remaining* (MOR) look instead at the job the operation belongs to, and favor the one with the largest amount of pending work, measured respectively in time units and in number of operations. The *critical ratio* (CR) weighs the work still to be done against the time available until a due date. Surveys catalog more than a hundred such variants [21, 22], and more recent ones compare those in current use across machine environments [23]. Beyond single-attribute rules, the algorithm of Giffler and Thompson [24], G&T in what follows, restricts each choice to a machine’s conflict set: the operations on the machine of the earliest-finishing candidate that can start before that candidate finishes. It guarantees *active* schedules and markedly improves plain rules when a priority criterion breaks the ties inside that set. We write G&T-SPT and G&T-MWKR for the resulting combinations. Stronger constructive procedures exist for the *deterministic* job shop, notably the shifting bottleneck heuristic [25], which repeatedly solves one-machine subproblems with heads and tails by means of Carlier’s algorithm [26]. These fall outside the scope of the present work: their machinery rests on exact one-machine optimization and critical-path arguments that do not extend directly to interval arithmetic, where no natural total order is available. No fixed combination of attributes performs well across shop configurations and objectives [22], which motivated the *automated* design of rules.

## 2.3. Automated design of dispatching rules

Genetic programming [27, 28] evolves a population of expressions, usually represented as trees, by applying selection and variation operators directly on their structure. GP hyper-heuristics apply this machinery to *priority expressions* over scheduling attributes, and are the standard approach to automated rule design for production scheduling [6, 7]. Evolved rules have been shown to outperform hand-crafted ones across job shop variants. Active research lines include surrogate fitness [29], feature selection [30] and multi-objective formulations [31]. Several evolved rules can also be combined into *ensembles*

that jointly construct or vote on a solution [32]. Systematic tree search over the space of expressions is an alternative to evolutionary search, and yields rules of comparable quality but smaller size and better readability [33]. All of this work targets deterministic or dynamic-stochastic settings. Closest to ours, Karunakaran et al. [34] evolve rules for dynamic job shops with uncertain processing times, exposing to the rule an exponential moving average of the observed deviations. Uncertainty there is realized stochastically during simulation, and rules are evaluated on sampled scenarios. The same holds for GP priority rules evolved for project scheduling with stochastic activity durations [35]. A recent survey of learned optimization for the job shop [36], covering both GP and DRL constructive approaches, records no work on interval-valued processing times. To the best of our knowledge, rules whose *inputs* are interval attributes, with fitness computed by interval arithmetic rather than by sampling realizations, have not been evolved before.

Deep reinforcement learning provides an alternative family of learned constructive methods for the JSP [37, 38]. GP rules and DRL policies are increasingly studied as competing paradigms for the same construction task, and controlled comparisons between them under common protocols have been identified as a gap in the recent literature [36]. At deployment both build a schedule in a single constructive pass. Han et al. [39] apply reinforcement learning to the IJSP itself. They learn which machine-order relations to fix before execution, and leave those most sensitive to the intervals to be resolved during execution by timed automata. On 14 classical instances they report lower mean midpoint makespans than population-based neighborhood search and a genetic algorithm. The rules evolved here, by contrast, fix every sequencing decision before execution.

### 3. The Interval Job Shop Scheduling Problem

An IJSP instance consists of  $n$  jobs and  $m$  machines. Job  $j$  is a chain of  $m$  operations  $o_{j1} \prec o_{j2} \prec \dots \prec o_{jm}$ , each requiring a distinct machine  $\mu(o_{jl})$  exclusively and non-preemptively. The duration of operation  $o$  is an interval  $\mathbf{p}_o = [\underline{p}_o, \bar{p}_o]$  with  $0 < \underline{p}_o \leq \bar{p}_o$ : the only assumption is that the realized duration lies within these bounds. Throughout, boldface denotes an interval-valued quantity, and underlined and overlined symbols its lower and upper bounds.

Schedule construction uses interval arithmetic. For intervals  $\mathbf{a} = [\underline{a}, \bar{a}]$

and  $\mathbf{b} = [\underline{b}, \bar{b}]$ , the two operations required by makespan minimization are

$$\mathbf{a} + \mathbf{b} = [\underline{a} + \underline{b}, \bar{a} + \bar{b}], \quad (1)$$

$$\max(\mathbf{a}, \mathbf{b}) = [\max(\underline{a}, \underline{b}), \max(\bar{a}, \bar{b})], \quad (2)$$

Both operations are component-wise, as is standard in interval arithmetic [40] and in the IJSP literature [2]. A solution determines an operation processing order  $\pi$  (a permutation with repetition of jobs, decoded left to right). Following the notation of Díaz et al. [2] and Díaz et al. [5], let  $\text{PM}_o(\pi)$  and  $\text{PJ}_o$  denote the predecessors of operation  $o$  in its machine (according to  $\pi$ ) and in its job. The *semi-active* schedule [41] starts each operation at

$$\mathbf{s}_o(\pi) = \max(\mathbf{c}_{\text{PJ}_o}(\pi), \mathbf{c}_{\text{PM}_o(\pi)}(\pi)), \quad \mathbf{c}_o(\pi) = \mathbf{s}_o(\pi) + \mathbf{p}_o, \quad (3)$$

so starting and completion times of all operations are themselves intervals. The problem can be stated as

$$\min_{\pi} \mathbf{C}_{\max}(\pi) \quad (4)$$

$$\text{s.t. } \mathbf{C}_{\max} = \max_{1 \leq j \leq n} \{\mathbf{c}_{o_{jm}}\}, \quad (5)$$

$$\underline{s}_{o_{jl}} \geq \underline{c}_{o_{j,l-1}}, \quad \bar{s}_{o_{jl}} \geq \bar{c}_{o_{j,l-1}}, \quad 2 \leq l \leq m, \quad 1 \leq j \leq n, \quad (6)$$

$$\underline{s}_o \geq \underline{c}_{o'} \vee \underline{s}_{o'} \geq \underline{c}_o, \quad \bar{s}_o \geq \bar{c}_{o'} \vee \bar{s}_{o'} \geq \bar{c}_o, \quad \forall o \neq o' : \mu(o) = \mu(o'), \quad (7)$$

where Eq. (5) defines the interval makespan as the component-wise maximum of the job completion times, Eq. (6) enforces precedence within each job, and Eq. (7) forbids overlaps on each machine in both interval components [2, 5]. Both components follow the same machine order, the one given by  $\pi$ . The minimum in (4) is taken over processing orders  $\pi$  with respect to an interval ranking  $R$ , since intervals admit no natural total order and any ranking embodies a modeling choice [42]. We rank schedules lexicographically, comparing upper bounds first and using the lower bound only to break ties:

$$\mathbf{a} \preceq \mathbf{b} \iff \bar{a} < \bar{b} \vee (\bar{a} = \bar{b} \wedge \underline{a} \leq \underline{b}), \quad (8)$$

This is one of the standard rankings of the IJSP literature, and the one that Díaz et al. [2] found to yield the most robust schedules. All methods in this paper use one implementation of this decoder and evaluator. The only exceptions are the two Giffler–Thompson baselines, which share the evaluator but restrict each choice to a conflict set (Section 2).

Performance is reported as the relative error of the expected makespan with respect to a crisp reference bound LB of each instance,

$$\text{RE} = \frac{E[\mathbf{C}_{\max}] - \text{LB}}{\text{LB}} \times 100, \quad E[\mathbf{C}_{\max}] = \frac{\underline{C}_{\max} + \overline{C}_{\max}}{2}, \quad (9)$$

where  $E[\mathbf{C}_{\max}]$  is the midpoint of the interval makespan. The IJSP literature calls it the expected makespan, because it is the expectation under a uniform distribution on  $[\underline{C}_{\max}, \overline{C}_{\max}]$ . Here it serves only as a point summary of the interval, and no distributional assumption enters the problem. LB is a bound of the underlying deterministic instance: Taillard’s bounds [43] for the Taillard-derived instances, and the best known ones [4] for the classical set. It is an indicative common reference, not a bound for the interval problem, for which none is available.

A schedule is eventually *executed* under one realization of the durations. A second quantity of interest is therefore how well its a-priori interval predicts that execution. Measuring the gap between a predicted schedule and its execution is the classical notion of schedule robustness [44], here specialized to interval predictions. Following Díaz et al. [2], the  $\bar{\varepsilon}$  robustness of a schedule with a fixed processing order is the average relative deviation of its executed makespan from the predicted expected value, over  $K$  realizations of the durations:

$$\bar{\varepsilon} = \frac{1}{K} \sum_{k=1}^K \frac{|C_{\max}^{\text{ex},k} - E[\mathbf{C}_{\max}]|}{E[\mathbf{C}_{\max}]}, \quad (10)$$

where  $C_{\max}^{\text{ex},k}$  is the crisp makespan obtained by executing that same order under the durations of realization  $k$ . A low  $\bar{\varepsilon}$  therefore means that the executed makespan stays close to its a-priori expectation.

## 4. GP Hyper-Heuristic for the IJSP

We describe the method from two complementary viewpoints: as a *hyper-heuristic*, that is, what an individual is and how it turns an IJSP instance into a schedule (Section 4.1); and as an *evolutionary algorithm*, that is, how the population of rules is initialized, varied and selected (Section 4.2).

### 4.1. Hyper-heuristic view: from expression trees to schedules

An individual is a priority expression tree (Figure 1). Leaves are drawn from the interval-aware terminal set of Table 1. Its three *width terminals*,

Table 1: Interval-aware terminal set. For an eligible operation  $o$  of job  $j$  on machine  $\mu(o)$ ,  $\mathbf{p}_o = [\underline{p}_o, \bar{p}_o]$  is its duration interval,  $\mathbf{C}_j$  and  $\mathbf{C}_{\mu(o)}$  are the interval completion times of the last scheduled operation of  $j$  and of  $\mu(o)$ ,  $\mathcal{R}_o$  is the set of operations of  $j$  after  $o$ , and  $\mathcal{E}$  is the eligible set. Terminals take raw, unnormalized values.

| Terminal | Definition                                                                                             | Role                            |
|----------|--------------------------------------------------------------------------------------------------------|---------------------------------|
| PT       | $\bar{p}_o$                                                                                            | worst-case processing time      |
| PTW      | $\bar{p}_o - \underline{p}_o$                                                                          | duration uncertainty            |
| EST      | $\bar{e}_o = \max(\bar{C}_j, \bar{C}_{\mu(o)})$                                                        | worst-case earliest start       |
| ESTW     | $\bar{e}_o - \underline{e}_o$ , with $\underline{e}_o = \max(\underline{C}_j, \underline{C}_{\mu(o)})$ | start uncertainty               |
| WKR      | $\sum_{q \in \mathcal{R}_o} \bar{p}_q$                                                                 | worst-case work remaining       |
| WKRW     | $\sum_{q \in \mathcal{R}_o} (\bar{p}_q - \underline{p}_q)$                                             | remaining-work uncertainty      |
| NOR      | $ \mathcal{R}_o $                                                                                      | operations remaining            |
| SLACK    | $\bar{e}_o - \min_{o' \in \mathcal{E}} \bar{e}_{o'}$                                                   | start slack within eligible set |
| ONE      | 1                                                                                                      | constant                        |

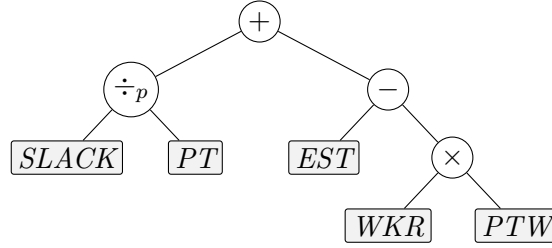


Figure 1: An individual is a priority expression tree; leaves are interval-aware attributes of the candidate operation (here the illustrative rule  $SLACK/PT + EST - WKR \cdot PTW$ ).

$PTW$ ,  $ESTW$  and  $WKRW$ , expose the widths of the intervals. Internal nodes are drawn from the function set

$$\mathcal{F} = \{ +, -, \times, \div_p, \min, \max, \text{neg} \}, \quad (11)$$

where neg is the unary sign change and  $\div_p$  denotes protected division, defined as  $a \div_p b = a/b$  if  $|b| > 10^{-9}$  and  $a$  otherwise.

The rule is deployed inside a semi-active schedule generation scheme (SGS). Starting from the empty schedule, the SGS takes  $nm$  decisions. At each one it forms the set  $\mathcal{E}$  of eligible operations, those whose job predecessor is already scheduled, and evaluates the tree on the attributes of every  $o \in \mathcal{E}$ . The operation with the lowest priority value is appended at its earliest feasible start on its machine, Eq. (3). Ties are broken by the lowest job index,

Table 2: Evolution parameters: the configuration selected by irace (Section 5.3), used for every experiment reported in this paper unless a section states otherwise.

|                            |                                                |
|----------------------------|------------------------------------------------|
| Population / generations   | 100 / 50                                       |
| Initialization             | ramped half-and-half (depths 2–4)              |
| Seeded individuals         | $PT$ and $\text{neg}(WKR)$                     |
| Selection                  | tournament, size $k = 7$                       |
| Crossover / mutation prob. | $p_c = 0.77 / 0.23$ (subtree, depth $\leq 3$ ) |
| Elitism                    | $e = 2$                                        |
| Tree-size cap              | $c = 30$ nodes (violations: fitness $\infty$ ) |
| Fitness                    | mean RE of one constructive pass, TA11–TA14    |
| Seeds                      | 1–30 (random generator of the evolution)       |

so a rule maps each instance to exactly one schedule.

Because the lowest value is dispatched, a criterion that seeks to *maximize* an attribute must be expressed through the unary  $\text{neg}$  operator. Classic rules thus have counterparts in this search space: the single-node tree  $PT$  is shortest processing time on the upper endpoints, and  $\text{neg}(WKR)$  is most work remaining. They differ from the SPT and MWKR baselines of Section 5.2 only on ties, which the trees break by job index and the baselines by the lower endpoint, Eq. (8). We inject both trees as seeded rules in the initial population.

#### 4.2. Evolutionary components

The initial population is built with the *ramped half-and-half* method [27, 28]. Each individual draws its maximum depth uniformly from  $\{2, 3, 4\}$ . With equal probability it is then generated by the *full* method, which expands every branch with function nodes down to that depth, or by the *grow* method, which may close a branch earlier by placing a terminal, with probability 0.3 per node. In both cases internal nodes are drawn uniformly from  $\mathcal{F}$  and leaves uniformly from the terminal set. The last two members of the population are not generated at random: they are the two seeded rules of Section 4.1, inserted directly. Every individual is evaluated by the mean RE of its deterministic constructive pass over the four training instances. The ranking of Eq. (8) compares schedules of a single instance, whereas fitness must summarize a rule over several, so it uses the scalar RE, the measure in which the IJSP literature reports its results.

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**Algorithm 1** GP hyper-heuristic for the IJSP

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1:  $P \leftarrow \text{ramped half-and-half}(\text{pop} - 2) \cup \{PT, \text{neg}(WKR)\}$ 
2:  $\text{evaluate}(P)$   $\triangleright$  mean RE of one constructive pass on the training
   instances
3: for  $g = 1$  to  $\text{gens}$  do
4:    $P' \leftarrow \text{elite}(P, e)$   $\triangleright$  elitism
5:   while  $|P'| < \text{pop}$  do
6:     if  $\text{rand}() < p_c$  then
7:        $p_1, p_2 \leftarrow \text{tournament}(P, k), \text{tournament}(P, k)$ 
8:        $\text{child} \leftarrow \text{subtreeCrossover}(p_1, p_2)$ 
9:     else
10:       $\text{child} \leftarrow \text{subtreeMutation}(\text{tournament}(P, k))$ 
11:    end if
12:     $P' \leftarrow P' \cup \{\text{child}\}$   $\triangleright$  fitness  $\infty$  if size  $> c$ 
13:  end while
14:   $P \leftarrow \text{evaluate}(P')$ 
15: end for
16: return best rule in  $P$ 
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Each generation then proceeds as follows. Parents are chosen by tournament, drawing  $k$  individuals uniformly without replacement from the current population and returning the fittest of them. With probability  $p_c$  two parents are recombined by *subtree crossover* [27, 28]. A node is drawn uniformly at random in each parent, leaves included. The offspring is a copy of the first parent in which the subtree rooted at its chosen node is replaced by the one rooted at the second parent's. Unlike the classical two-offspring formulation, we keep a single offspring per recombination. With the complementary probability, a single parent undergoes *subtree mutation*: a node is drawn uniformly at random, and the subtree rooted at it is replaced by a random tree built with the grow method and maximum depth 3. Offspring whose size exceeds a cap of  $c$  nodes receive infinite fitness and are thereby excluded from selection. The depth bounds apply only to trees created from scratch, at initialization and inside mutation. During the run only the node count is bounded, so recombination may produce deeper expressions.

Replacement is fully generational, the offspring becoming the next population, except for the  $e$  best individuals of the current generation, which are copied into it unchanged. The process stops after a fixed number of

generations and returns the best rule found. Algorithm 1 summarizes the generational loop and Table 2 lists the parameter values.

#### 4.3. Randomized multi-start extension

A deterministic rule produces one schedule. For comparisons with sampling-based methods at matched budget, we wrap the evolved rule in GRASP-style randomization [45]. With probability 0.1 the dispatched operation is drawn uniformly from the eligible set; otherwise the rule decides. Sample 0 remains deterministic. The wrapper, which we call GP-best-of- $N$ , reports the lexicographically best of its  $N$  schedules. This uniform perturbation of a single rule is the simplest member of a family of randomized rule-based builders; an alternative swaps among a *set* of evolved rules during construction [46]. GP-best-of- $N$  turns any budget beyond a single pass into better schedules. As in GRASP, the gain comes from sampling more schedules, not from better individual decisions. It is compared with per-instance search in Sections 6.4 and 6.5.

## 5. Experimental Setup

### 5.1. Benchmark instances and data split

We use two benchmarks from the IJSP literature [2, 4, 5]. The first is the 70 interval versions of Taillard’s instances [43] (TA1–TA70), in seven size classes from  $15 \times 15$  to  $50 \times 20$ . The second is the 12 classical instances on which the published IJSP metaheuristics report results: FT10 and FT20 [47], seven instances of the La family [48] and ABZ7–9 [25], with 10 to 20 jobs and 5 to 15 machines. Both were obtained from their crisp counterparts with the generation scheme of Díaz et al. [2]. For every operation of original duration  $p$ , a value  $\delta$  is drawn uniformly at random in  $[0, 0.15p]$ , and the duration becomes the integer interval  $[p-\delta, p+\delta]$ . Machine routes are unchanged. The interval is symmetric around  $p$ , so the crisp instance is exactly the midpoint scenario of its interval version. Both sets are released with this paper (see Data availability).

The 70 interval Taillard instances are split three ways. Rules are evolved on TA11–TA14 ( $20 \times 15$ ). TA15–TA20 form the validation set: they never contribute to fitness, but the design and configuration decisions of this paper, including the irace study of Section 5.3, were selected on their performance. The remaining 60 instances play no part in any of these decisions. Together

with the 12 classical instances, to which the rules are applied without re-evolution (Section 6.3), they form the generalization test.

Thirty independent evolutions, differing only in their random seed, yield thirty rules. The one with the lowest mean RE on the validation set is the *featured rule*. Results are reported as the mean over the 30 rules, which involves no selection, together with the value of the featured rule. Where a section examines a single rule, it states how that rule is chosen.

For the analysis of Section 7.7, the 70 Taillard instances were also re-generated with intervals skewed to the right. The crisp duration  $p$  is recovered from the midpoint of the symmetric version, which the scheme above preserves exactly. The bounds are drawn as  $p - \text{round}(pU[0, 0.05])$  and  $p + \text{round}(pU[0, 0.25])$ , so a duration may fall up to 5% below its nominal value and up to 25% above it. Machine routes are unchanged, and the crisp Taillard bounds still apply. Over the 37,250 operations, the mean width is 14.3% of  $p$ , against 12.9% for the symmetric scheme. The two benchmarks thus differ above all in the shape of the intervals; the asymmetric ones are also wider, by 1.4 points of  $p$  on average.

## 5.2. Reference methods

The constructive baselines are the dispatching rules of Section 2 (SPT, LPT, EST, MWKR, MOR and CR) and the two Giffler–Thompson combinations, G&T-SPT and G&T-MWKR, each applied in one deterministic pass. Each ranks candidates by its own criterion under the lexicographic order of Eq. (8). Section 7.3 checks that this choice does not handicap them. A dispatcher that chooses uniformly among the eligible operations serves as a reference point.

The sampled variant of the evolved rule is GP-best-of- $N$  (Section 4.3). The published IJSP metaheuristics are implemented in C++ and report results under their own stopping rules, which does not allow them to be compared with the evolved rule at a matched budget. For that comparison we implemented our own version of the genetic algorithm of Díaz et al. [2] in Python, on the decoder and evaluator of Section 3, so that it and the evolved rule differ only in the method. A solution is a permutation with repetition of jobs, decoded by the semi-active scheme of Eq. (3) and compared by the ranking of Eq. (8). A population of 250 individuals is initialized at random. Parents are chosen by tournaments of size 3 and recombined with probability 0.9 by a job-based order crossover: a random subset of jobs keeps its positions from the first parent, and the remaining positions are filled in the

Table 3: Hyperparameter space explored by irace (200 experiments; cost = validation-set RE of the evolved rule) and the winning configuration.

| Parameter             | Range / values       | irace winner |
|-----------------------|----------------------|--------------|
| Tournament size       | $\{2, 3, 4, 5, 7\}$  | 7            |
| Crossover prob. $p_c$ | $[0.5, 0.95]$        | 0.77         |
| Tree-size cap         | $\{20, 30, 40, 60\}$ | 30           |
| Elitism               | $\{1, 2, 4\}$        | 2            |

order they take in the second. Offspring are mutated with probability 0.2 by swapping two positions, and the two best individuals pass unchanged to the next generation.

### 5.3. Hyperparameter configuration with irace

Four parameters of the evolution were configured with irace [49]: tournament size, crossover probability, tree-size cap and elitism. Population size and number of generations were held fixed, so every configuration runs the same number of evaluations. irace was given 200 experiments, each a full evolution on the training instances scored by the validation RE of the resulting rule. Table 3 lists the parameter space and the winning configuration (Table 2). Of the four elite configurations that irace retains, three favor a high selection pressure (tournament size 7), and all four fall in a narrow crossover band, 0.72–0.77. The tree-size cap does not discriminate: the elites span caps of 20, 30 and 40.

### 5.4. Evaluation measures and statistical analysis

Every measure below is to be minimized. Those that concern the execution of a schedule are estimated by Monte Carlo over realizations of the durations, as described in Section 7.6.

- *Quality*: the relative error RE of Section 3, in percent.
- *Width of the prediction*: the relative width of the interval makespan,  $W = (\bar{C}_{\max} - \underline{C}_{\max})/E[\mathbf{C}_{\max}] \times 100$ .
- *Executional robustness*: the mean deviation of the executed makespan from its prediction, relative to the expected makespan ( $\bar{\epsilon}$ , Eq. (10)) and in time units ( $|\Delta|$ , the same average before dividing by  $E[\mathbf{C}_{\max}]$ ).

- *Tail of the execution*: the conditional value-at-risk at level 0.95,  $\text{CVaR}_{0.95}$ , the mean of the 5% largest values of a quantity over the same realizations. It is applied to the overrun  $C_{\max}^{\text{ex}} - E[\mathbf{C}_{\max}]$  of the executed makespan beyond its prediction, in time units, and to the executed makespan itself, expressed as RE over the same bound LB.

Comparisons are of two kinds, with different experimental units. Methods are compared instance by instance, pairing their values on the same instances; when a method is run several times on each instance, its value on an instance is the mean over the runs. The test is the two-sided Wilcoxon signed-rank test with exact  $p$ -values, reported with the normal-approximation statistic  $z$  and, as effect size, the matched-pairs rank-biserial correlation  $|r|$ . An *arm* is a set of evolutions, 30 unless stated otherwise, run with the same configuration (terminal set, fitness and training instances) and differing only in the random seed. The *full* arm uses the complete terminal set, and the *ablated* one omits the width terminals. Two arms share nothing but the seed numbers, so they are compared as independent samples of rules, each rule averaged over the instances. The test is the two-sided Mann–Whitney rank-sum test, reported with its normal-approximation  $z$ , negative when the first arm named has the lower values. Its effect size is the rank-biserial correlation  $|r|$ , which is 1 when the two arms do not overlap. In Section 6.5, whose many contrasts are each over 10 to 12 instances, the Wilcoxon test is reported instead with the number of instances favoring each method. Holm’s correction is applied within each family of related comparisons, and where it changes a verdict the text says so.

### 5.5. Computing environment

All experiments ran on a desktop computer with an AMD Ryzen 5 3400G processor (4 cores, 8 threads) and 32 GB of RAM under Windows 10 Pro. The hyper-heuristic, the decoder, the baselines and the genetic algorithm are implemented in Python 3.12 with NumPy 2.3, statistical tests use SciPy 1.17, and the configuration study used irace 4.4.3 [49]. All reported times are wall-clock times of a single simulator, which builds exactly the same schedules as the one used during evolution. They were measured with six runs executing in parallel, and are comparable with one another but not with times obtained on other hardware or in other languages.

## 6. Results

### 6.1. Evolution behavior and seed stability

Over 30 independent evolutions, training RE converges to  $14.9 \pm 1.5$  (range 12.4–19.7), with most of the improvement realized by generation 25–30 (Figure 2a). Each evolution takes about 26 min (24–28 over the 30 seeds) on the machine of Section 5.5, an offline cost paid once per rule. The same panel evaluates the best rule of each generation on the instances the fitness never sees, and there the improvement is concentrated in the first ten generations. From generation 10 to 50 the median training RE falls from 17.09 to 14.57. On the 60 test instances it falls only from 19.32 to 18.71, and on the validation set it hardly moves, from 19.85 to 19.69. The later generations thus specialize the rule to the four training instances more than they improve it elsewhere, and the validation set could also serve to stop the evolution earlier at a small cost in quality. Over the same generations the best rules grow to a median of 27 nodes, just below the cap of 30 (Figure 2b). Interval-width terminals make up about a fifth of their leaves throughout (Figure 2c). Figure 3 shows the featured rule, obtained with seed 1: 26 nodes over 12 leaves. Every terminal of the tree is non-negative by construction and processing times are at least one time unit, so its min and max nodes always select the same argument. On the problem domain the tree is therefore exactly equivalent to the expression

$$SLACK^2 + 2PT - WKR - WKRW - 1. \quad (12)$$

We verified this identity at each of the 37,250 dispatching decisions the rule takes on the benchmark (1.29 million candidate evaluations).

### 6.2. Position among constructive methods

Table 4 reports every constructive baseline under the protocol of this paper, one deterministic pass per instance, together with the random dispatcher of Section 5.2. Since these rules are deterministic, the standard deviation is taken across instances: it measures how uniform a method is over the benchmark, not any run-to-run variability. The random dispatcher, the one stochastic entry, is averaged over 10 dispatches per instance. Times are averages over the 70 instances with 5 repetitions each. The GP entries are the featured rule, timed as evolved, and the same rule in its simplified form, which takes the same decisions.

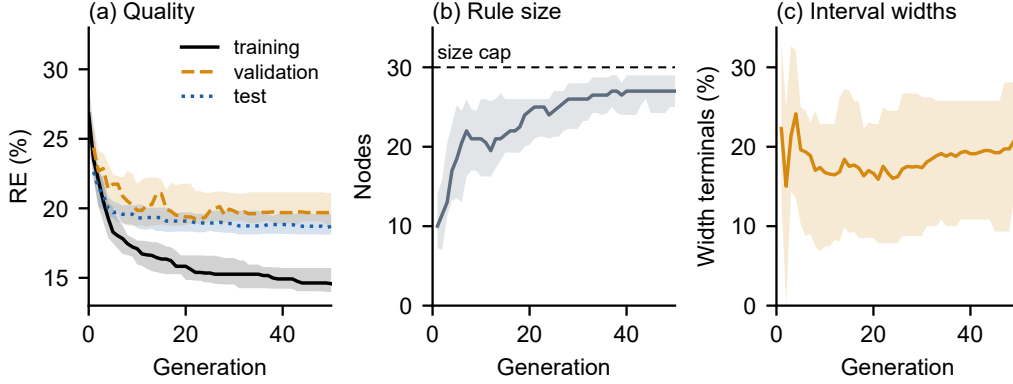


Figure 2: The best rule of each generation over the 30 evolutions: (a) RE on the training, validation and test instances, (b) size in nodes, (c) share of interval-width terminals among its leaves. Lines are medians (means in c); bands are interquartile ranges.

Two observations are needed to read the table correctly. First, the plain rules that ignore machine availability, SPT, LPT and the critical ratio CR, perform *worse than random* in this setting; CR is a due-date rule, applied here without due dates. This is a property of the schedule generation scheme, not a defect of the criteria: because a solution is decoded as a permutation whose operations are appended to their machines, systematically dispatching by processing time alone produces highly unbalanced permutations. The same criterion embedded in the Giffler–Thompson conflict set, which restores classical dispatching semantics, behaves as expected (SPT: 627% as a plain rule versus 70.6% under G&T). Second, the criteria that do account for the state of the shop (EST, MOR, MWKR) are better than random, and the strongest baseline overall is G&T-MWKR at 29.5%, which is the reference we use throughout.

Against that strongest baseline, the 30 evolved rules average  $18.99 \pm 1.33$  RE, and the featured rule attains 17.71%, the lowest of the 30 on these instances although it was selected on the validation set.

The gap holds on every instance: the featured rule outperforms each of the eight deterministic baselines on all 70, and a paired Wilcoxon signed-rank test rejects equality against each of them at  $p < 10^{-6}$  ( $z = -7.3$ ), also after Holm’s correction for the eight comparisons. The narrowest margin over the whole comparison is 3.9 points, against G&T-MWKR on a single  $20 \times 15$  instance. Against the best plain rule, EST, the smallest margin is

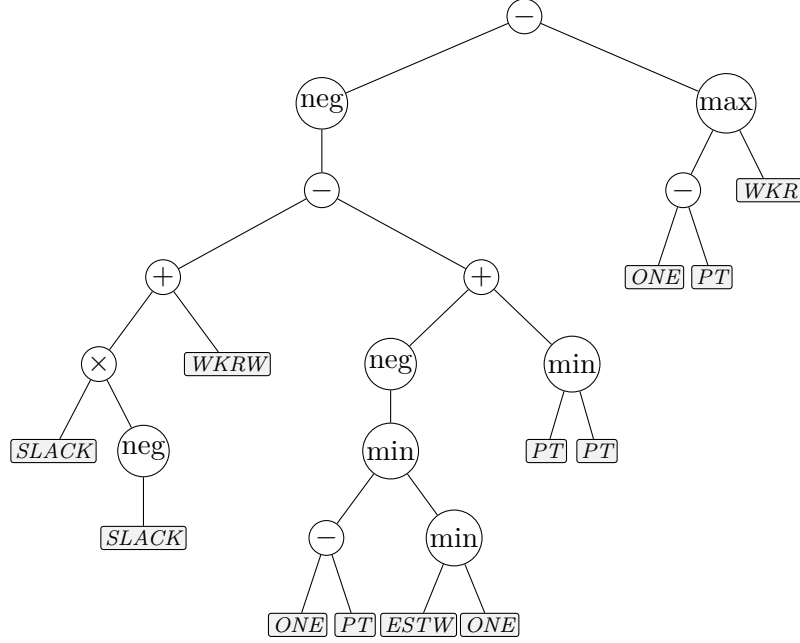


Figure 3: The featured rule (seed 1) as evolved; shaded boxes are the terminals of Table 1, and Eq. (12) gives its simplified form.

6.3 points. Per-instance figures are given in Supplementary Table S3. This quality comes at a small evaluation cost. A pass of the featured rule takes 20.6 ms as evolved and 11.1 ms in its simplified form, against 17.4 ms for G&T-MWKR, the strongest baseline, and 2.7 ms for MOR, which reads a single attribute.

### 6.3. Generalization across sizes and instance families

Table 5 reports single-pass RE per size class over all 70 instances, as mean  $\pm$  standard deviation across the 30 evolved rules and for the featured rule. Evolved only on  $20 \times 15$ , the featured rule attains its lowest RE on the unseen  $50 \times 15$  (13.8%), and the same pattern holds on average ( $13.7 \pm 1.5$ ). Because all terminals are per-operation attributes with no size-bound quantities, transfer requires no retraining or rescaling.

The ranking of the classes reflects the difficulty of their instances more than their distance from the training size. In the two 50-job classes, where the rule does best, every crisp counterpart has been solved to optimality [50]. The bounds used here equal the optimal makespans, so on these classes RE

Table 4: Constructive baselines on the 70 interval Taillard instances: mean RE (%), standard deviation across instances, and time of one pass (ms), all on the same simulator.

| Method                   | RE (%)       | sd          | Time (ms)  |
|--------------------------|--------------|-------------|------------|
| LPT                      | 703.6        | 168.0       | 4.2        |
| SPT                      | 627.2        | 152.2       | 4.2        |
| CR                       | 625.7        | 140.6       | 3.3        |
| <i>Random dispatcher</i> | <i>127.2</i> | <i>13.9</i> | <i>1.1</i> |
| G&T-SPT                  | 70.6         | 10.0        | 14.3       |
| MWKR                     | 64.7         | 10.4        | 5.0        |
| MOR                      | 45.5         | 10.1        | 2.7        |
| EST                      | 42.3         | 10.6        | 9.8        |
| G&T-MWKR                 | <b>29.5</b>  | 6.4         | 17.4       |
| GP rule (featured)       | <b>17.71</b> | 5.23        | 20.6       |
| simplified, Eq. (12)     | 17.71        | 5.23        | 11.1       |

Table 5: RE (%) per size class over the 70 interval Taillard instances: mean  $\pm$  sd of the 30 evolved rules, and the featured rule.

|               | 15 $\times$ 15 | 20 $\times$ 15 | 20 $\times$ 20 | 30 $\times$ 15 | 30 $\times$ 20 | 50 $\times$ 15 | 50 $\times$ 20 | All              |
|---------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|------------------|
| Mean of 30    | 18.1 $\pm$ 1.3 | 18.0 $\pm$ 1.5 | 20.9 $\pm$ 1.7 | 21.4 $\pm$ 1.7 | 25.7 $\pm$ 1.9 | 13.7 $\pm$ 1.5 | 15.1 $\pm$ 1.5 | 18.99 $\pm$ 1.33 |
| Featured rule | 15.7           | 16.6           | 18.1           | 21.1           | 24.1           | <b>13.8</b>    | 14.6           | <b>17.71</b>     |

is measured against the optimum. The error is largest on 30 $\times$ 20, the class with the most open instances, nine of ten. The reference bound of an open instance lies below its unknown optimum, so part of the error measured there belongs to the bound: the best known solutions of that class exceed the bounds used here by 2.7% on average.

The rules also transfer to the second benchmark, whose 12 classical instances differ from the Taillard ones in origin, size and structure. Table 6 gives the featured rule’s single-pass RE on each, together with GP-best-of- $N$  at  $N = 64$  and  $N = 1024$  (GP-64 and GP-1024) and the published results of the IJSP metaheuristics, which Section 6.4 discusses.

Transfer is a property of the method rather than of one rule: over the 30 evolved rules the deterministic pass averages  $19.04 \pm 1.97\%$  RE on these instances, single rules ranging from 15.3% to 23.3%. Since the two benchmarks use different reference bounds (Section 3), RE values do not compare

Table 6: RE (%) on the 12 classical interval instances: our methods and the published averages over 30 runs per instance of GA [2],  $ABC_{E3}$  [3], fEABC [4] and ESABC [5]. LB is the reference bound of Section 3; GP-64 and GP-1024 are GP-best-of- $N$  with  $N = 64$  and 1024.

| Inst. | LB   | Ours (zero-shot) |       |         | Published (avg of 30) |            |       |       |
|-------|------|------------------|-------|---------|-----------------------|------------|-------|-------|
|       |      | GP               | GP-64 | GP-1024 | GA                    | $ABC_{E3}$ | fEABC | ESABC |
| FT10  | 930  | 20.4             | 8.4   | 6.6     | 5.2                   | 4.1        | 3.5   | 3.0   |
| FT20  | 1165 | 13.2             | 12.1  | 10.8    | 4.4                   | 1.7        | 1.8   | 1.8   |
| La21  | 1046 | 15.0             | 14.4  | 9.0     | 5.0                   | 5.0        | 4.2   | 4.0   |
| La24  | 935  | 10.1             | 8.8   | 8.6     | 6.3                   | 5.1        | 4.9   | 5.0   |
| La25  | 977  | 11.8             | 11.6  | 7.6     | 5.1                   | 3.9        | 3.4   | 2.7   |
| La27  | 1235 | 25.0             | 13.7  | 11.3    | 10.2                  | 4.7        | 4.6   | 4.1   |
| La29  | 1152 | 13.4             | 14.2  | 11.0    | 14.2                  | 8.6        | 7.4   | 7.0   |
| La38  | 1196 | 15.0             | 13.3  | 9.4     | 9.2                   | 6.9        | 6.1   | 5.8   |
| La40  | 1222 | 13.3             | 12.8  | 9.9     | 8.7                   | 4.2        | 4.6   | 4.1   |
| ABZ7  | 656  | 19.7             | 14.3  | 11.7    | 12.5                  | 7.3        | 6.6   | 6.7   |
| ABZ8  | 645  | 24.7             | 22.1  | 17.7    | 18.5                  | 12.1       | 11.1  | 10.9  |
| ABZ9  | 661  | 33.4             | 25.3  | 22.1    | 18.0                  | 13.1       | 11.6  | 11.2  |
| Mean  |      | 17.9             | 14.2  | 11.3    | 9.8                   | 6.4        | 5.8   | 5.5   |

across them, but margins over baselines evaluated on the same instances do. That margin is preserved: 2.40 times over MOR and 1.58 over G&T-MWKR on the classical set, against 2.40 and 1.55 on the benchmark the rules were evolved for.

#### 6.4. Comparison with published metaheuristics

On the 12 classical instances the evolved rules can be compared with the published results of the IJSP metaheuristics on their own benchmark and metric (Table 6). Those methods belong to different algorithmic families, evolutionary computation and swarm intelligence: a genetic algorithm evolving a population of permutations [2] and three artificial bee colony variants [3–5], each of which runs a separate search on every instance.

In a single pass the evolved rules average 19.04% on these instances, against 30.1% (G&T-MWKR) and 45.8% (MOR) for the deterministic baselines. With a 1024-sample budget the featured rule reaches 11.3%, within 1.5 points of the 9.8% published for the genetic algorithm, which searches every

instance separately. It is better than the genetic algorithm on 3 of the 12 instances (paired Wilcoxon  $z = 1.96$ ,  $p \approx 0.05$ ). The ABC variants, which add neighborhood structures and local search to the population, remain ahead (5.5–6.4%), as expected of purpose-built iterative search. The comparison thus places the evolved rules at the frontier of constructive methods, close to a plain population-based search.

The two families also differ in how they spend computation: GP-1024 evaluates 1024 mutually independent samples per instance, which can run in parallel, whereas the metaheuristics evolve a population of 250 individuals [2, 4] through successive generations, each of which depends on the previous one.

#### *6.5. Comparison at matched budget with a reimplemented genetic algorithm*

For a comparison at matched budget we implement the genetic algorithm of Díaz et al. [2] (Section 5.2) in Python; its implementation is included in the data repository (see Data availability). On the 12 classical instances it achieves 7.3% RE in 900 s per instance, against the 9.8% published for the original, whose runs are shorter. The rule and the genetic algorithm share decoder, evaluator and language. The seeded genetic algorithm places the permutation built by one pass of the featured rule in its initial population, alongside 249 random ones. The comparison is run on the 12 classical instances, with 30 runs of 900 s per instance and method, and on three Tailard classes representative of the different instance sizes,  $15 \times 15$ ,  $30 \times 15$  and  $50 \times 15$ . Each method is run 3 times on each of the 10 instances of a class, for a total of 30 runs per class. The horizons are 900, 1800 and 3600 s per instance, chosen to extend well beyond the point where the curves converge (Figure 4).

At low budgets GP-best-of- $N$  attains the lowest RE of all methods, until the genetic algorithm surpasses it. The budget at which this occurs increases with instance size, from approximately 10 s per instance on the classical instances and 25 s on  $15 \times 15$  to approximately 300 s on  $50 \times 15$ , with 83 s on the intermediate  $30 \times 15$  class. The rule is therefore the most effective method at low budgets, a regime that extends further the larger the instance. Seeding the genetic algorithm with the featured rule likewise yields an initial advantage over the unseeded version, but the two converge: from 100 s per instance onwards, the statistical tests detect no difference between them on any instance set (Supplementary Table S4). An evolved dispatching rule thus delivers in a single pass of milliseconds a quality that the genetic algorithm needs between 2 and 46 s per instance to reach. As a seed, it gives

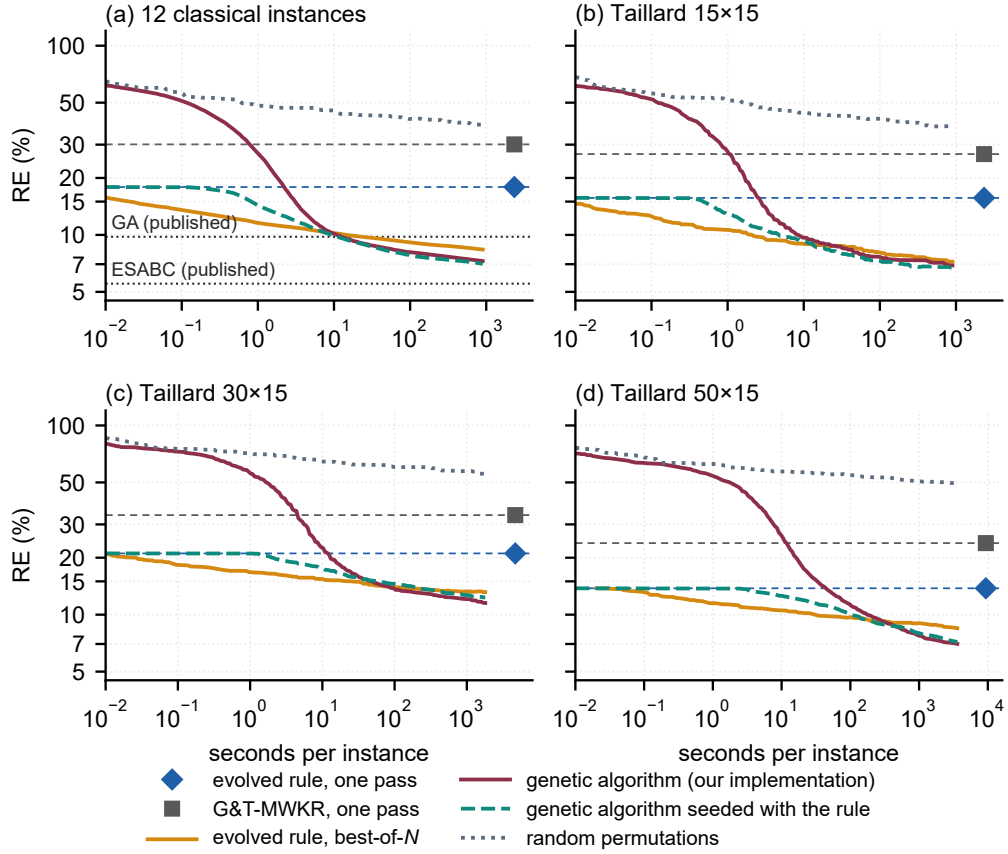


Figure 4: Mean RE of the best schedule found, over runs and instances, against seconds per instance on the 12 classical instances (a) and on three Taillard classes with 15 machines, 15×15 (b), 30×15 (c) and 50×15 (d).

a population-based method a high-quality initial solution from which the search progresses faster, the role constructive heuristics commonly play in the initialization of such methods [19].

## 7. Analysis

### 7.1. Rule anatomy and interpretability

The output of GP is a readable expression. Evolved rules are compact, with a mean tree size of 26 nodes (20–30) and mean depth 9, so a rule can be inspected and simplified by hand. Figure 5 reports how often each

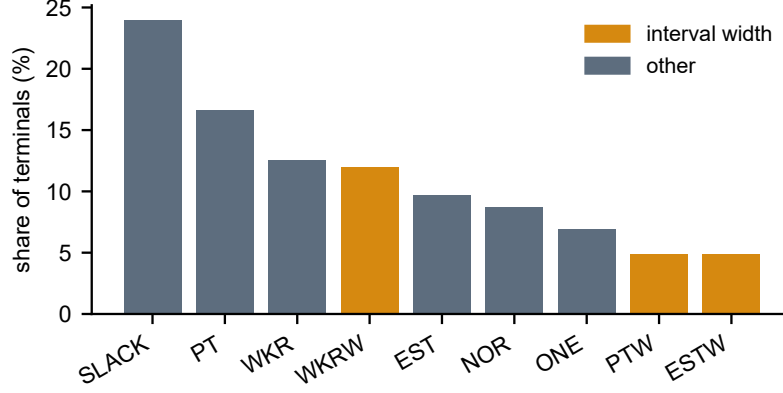


Figure 5: Terminal usage across the 30 evolved rules, as a share of all terminals; interval-width terminals in amber.



Figure 6: Mean RE of the featured rule over the 70 instances as each weight of its simplified form, Eq. (12), varies with the other fixed. Dashed lines: evolved weights; dotted line: mean of the 30 evolved rules;  $\beta = 0$  removes the width term.

terminal is used across the 30 evolved rules. The most frequent are the classical scheduling attributes, slack (*SLACK*), work remaining (*WKR*) and earliest start (*EST*), together with worst-case processing time (*PT*): the evolution builds on known dispatching logic. The *interval-width* terminals (*PTW*, *ESTW*, *WKRW*) account for 21.7% of all terminal usage and appear in 27 of the 30 evolved rules. Whether that width information translates into

better schedules is quantified in Sections 7.4–7.6.

Since the lowest value is dispatched, the featured rule, Eq. (12), reads directly:  $-WKR$  is most-work-remaining,  $2PT$  is shortest-processing-time at double weight,  $SLACK^2$  penalizes quadratically any operation that cannot start at the earliest available time, and the constant is immaterial to the ranking. The remaining term,  $-WKRW$ , is the interval-aware one. Among jobs with comparable remaining work, the rule advances those whose remaining work is *more uncertain*, so that this uncertainty is resolved early rather than carried to the end of the schedule; the worked example of Section 7.2 illustrates it. The evolved rule is thus a weighted blend of MWKR and SPT, regularized by a quadratic earliest-start term and tilted by the uncertainty of the work ahead.

The simplified form of the featured rule, Eq. (12), also admits a direct sensitivity analysis: it weighs  $PT$  by  $\alpha = 2$  and  $WKRW$  by  $\beta = 1$ , and we swept each weight over the 70 instances holding the other at its evolved value (Figure 6). Neither curve is unimodal: over the sampled grid,  $\alpha$  has five local minima and  $\beta$  three. Each point is a deterministic evaluation over the 70 instances, so this ruggedness belongs to the landscape, not to sampling noise. The evolved  $\alpha = 2$  is the global minimum of its sweep. The evolved  $\beta$  is not: it gives 17.71%, against 17.54% for  $\beta = 0.5$ . A post-evolution weight sweep on the validation set is thus a cheap refinement for any simplified rule. The multimodality becomes relevant if the sweep is replaced by local descent: started at  $\alpha = 4$ , a descent converges to a secondary minimum 0.6 points above the global one. Setting  $\beta = 0$  removes the width term from the deployed rule and costs 0.5 points of RE (18.23 against 17.71).

## 7.2. A worked example

The example compares the rule with its variant without the width term ( $\beta = 0$  in Section 7.1) on a small instance with three jobs of three operations.  $J_1$  visits  $M_3, M_2, M_1$  with durations  $[28, 30], [20, 26], [24, 32]$ ;  $J_2$  visits  $M_1, M_3, M_2$  with  $[26, 30], [10, 12], [16, 18]$ ; and  $J_3$  visits  $M_3, M_1, M_2$  with  $[22, 26], [14, 14], [27, 31]$ .

The two traces, that is, the sequences of dispatching decisions, coincide in the first two decisions. They diverge at the third, where  $M_3$  has just been released by  $o_{11}$  and the eligible set is  $\{o_{12}, o_{22}, o_{31}\}$ . Table 7 gives the terminals and the resulting scores. All three candidates can start at the same time, so  $SLACK$  vanishes and the decision rests on the remaining terms. Without the width term the ranking puts  $o_{22}$  (5) ahead of  $o_{31}$  (6): the short

Table 7: Terminals and priority scores at the third dispatching decision of the worked example.

| Eligible | Duration | WKR | WKRW | SLACK | Eq. (12) | $\beta = 0$ |
|----------|----------|-----|------|-------|----------|-------------|
| $o_{12}$ | [20, 26] | 32  | 8    | 0     | 11       | 19          |
| $o_{22}$ | [10, 12] | 18  | 2    | 0     | 3        | <u>5</u>    |
| $o_{31}$ | [22, 26] | 45  | 4    | 0     | <u>2</u> | 6           |

operation of the job with little work remaining ranks first because of the 2  $PT$  term. The width term contributes  $-WKRW$ , which is  $-4$  for  $o_{31}$  against  $-2$  for  $o_{22}$ , and that reverses a one-point gap. The rule therefore dispatches  $o_{31}$ , whose job has more uncertain remaining work than that of  $o_{22}$ , [41, 45] against [16, 18]. In this same state, SPT would dispatch  $o_{22}$  and MWKR  $o_{31}$ .

Figure 7 shows what follows. Dispatching  $J_3$  early schedules its remaining operations while the machines are still available, and the makespan is  $C_{\max} = [103, 117]$ . Delaying it by that single decision postpones the same operations to the end of the schedule, and the makespan becomes [113, 133], worse at both endpoints. Run on their own from the empty schedule (Figure 7c and d), SPT and MWKR yield [147, 170] and [143, 161], against a lower bound of [72, 88] set by the total work of  $J_1$ . Neither looks at when a candidate can start, and the decoder appends each operation after the last one on its machine. MWKR, which already departs from the evolved rule at the second decision, dispatches  $o_{32}$  before  $o_{21}$  and leaves  $M_1$  idle until [50, 56]. As a result  $o_{21}$ , which could have started at time 0, starts at [64, 70]. SPT leaves  $M_3$  idle from [22, 26] to [62, 70] in the same way. The  $SLACK^2$  term of the evolved rule penalizes a candidate by how much later it would start than the earliest one.

The example illustrates the mechanism, not its magnitude. Over 800 random  $3 \times 3$  instances generated with the scheme of Section 5.1, removing the width term leaves the trace unchanged in 668. Of the 132 traces that do change, 93 end at the same makespan, 21 end better with the term and 18 worse. Its contribution over the 70-instance benchmark is measured in Sections 7.4 and 7.6.

### 7.3. Interval handling and the decoder

The comparison with the constructive baselines could depend on two choices that are not part of the rules themselves, and Supplementary Sec-

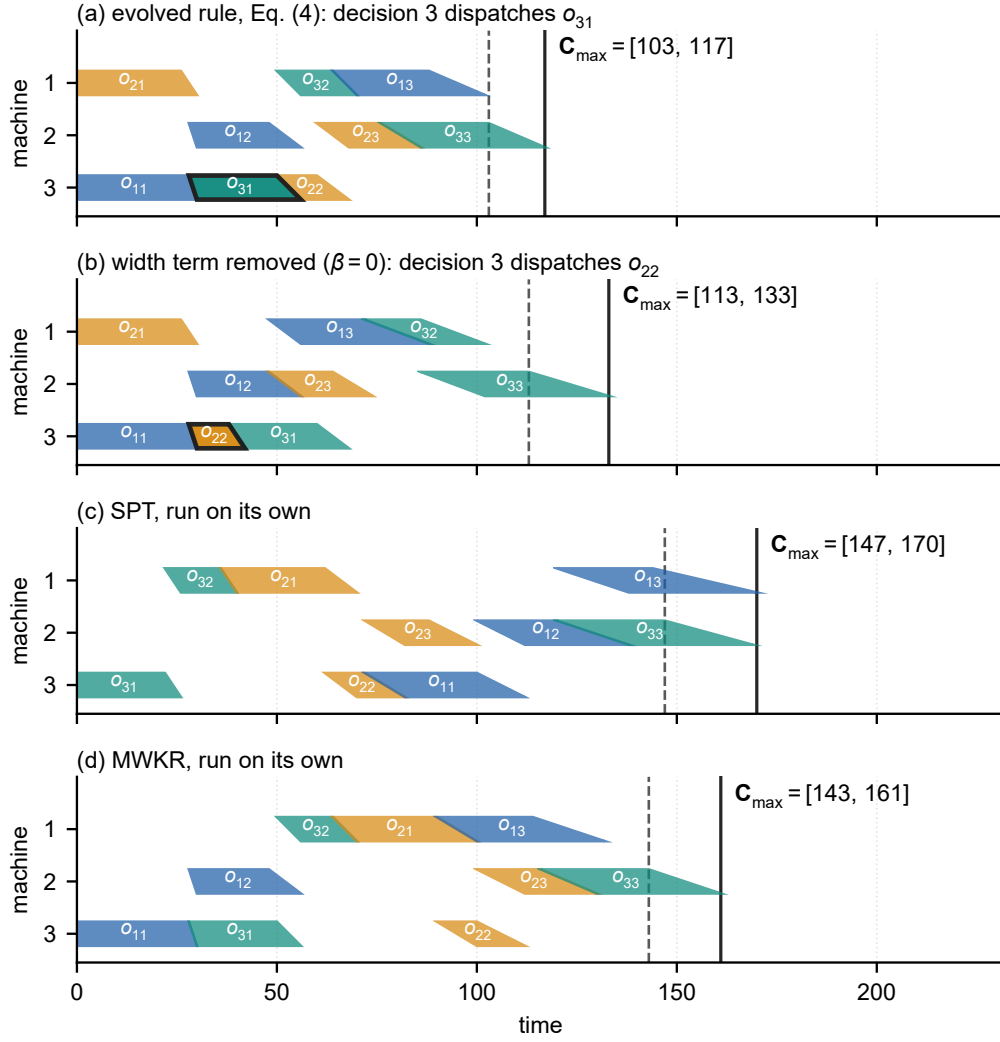


Figure 7: The worked example scheduled by (a) the evolved rule, (b) the rule without the width term, (c) SPT and (d) MWKR. The outlined operation separates traces (a) and (b).

tion S1 examines both. The first is how a baseline compares intervals. Table 4 ranks candidates lexicographically, Eq. (8); ranking them instead by the lower endpoint, the midpoint or the upper endpoint changes the RE of EST and MWKR by at most 2.4 points. That is far less than the 23.3 points that separate the best of them, EST, from the evolved rules. The second is the decoder. The evolved rules choose among all eligible operations and build

semi-active schedules, whereas G&T-MWKR, the strongest baseline, chooses only within Giffler and Thompson’s conflict set and builds active ones. Exchanging the decoders shows that the evolved rules outperform MWKR under both: 18.99 against 64.7 with the semi-active decoder and 23.18 against 29.5 with the conflict set. Each method in Table 4 is therefore reported with the decoder that suits it, and the margin of the evolved rules does not depend on that choice.

#### 7.4. The value of interval awareness: ablations

The interval-width terminals are the distinctive ingredient of this setting, so we isolate their contribution by ablation. The 30 evolutions are repeated without the width terminals (*PTW*, *ESTW*, *WKRW*), with everything else held fixed, including the irace configuration of Section 5.3. Interval information also enters the evolution through the fitness, which scores every schedule by interval arithmetic. To remove it as well, 30 further rules are evolved on the crisp midpoint counterparts of the training instances (Section 5.1), with the same configuration and reduced terminal set, so that they never handle interval data. Figure 8 shows the 30 rules of each arm, compared as independent samples (Section 5.4). Supplementary Table S5 gives their means; Holm’s correction over its tests changes none of the verdicts.

The width terminals change the schedules even when nothing rewards it: under the makespan objective, the rules that read them predict narrower makespan intervals ( $z = -2.96$ ,  $p < 0.01$ ,  $|r| = 0.44$ ). In expected makespan, the quantity this objective optimizes, the evolution reaches the same quality with or without them (medians 18.52 and 18.61;  $z = 0.67$ , n.s.). The crisp midpoint control matches the interval-trained arm on both measures ( $z = -0.12$  and  $z = -1.26$ , n.s.), although the two fitness functions differ, since max does not commute with taking midpoints. What interval awareness adds is therefore not a lower expected makespan but control over the width of the prediction, which Section 7.5 examines under an objective that rewards it.

#### 7.5. The robust objective: trading expected makespan for predictability

The makespan fitness gives the evolution no incentive to trade expected makespan for predictability. To test whether it can make that trade when rewarded for it, we repeat the terminal ablation with a *robust* fitness that explicitly rewards predictability, minimizing

$$f_\lambda = \overline{C}_{\max} + \lambda (\overline{C}_{\max} - \underline{C}_{\max}). \quad (13)$$

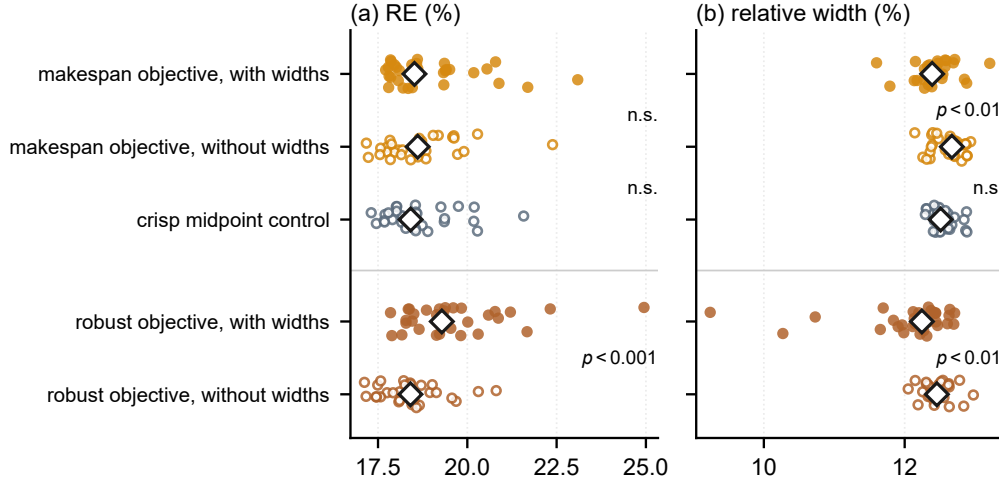


Figure 8: The 30 evolved rules of each ablation arm, each averaged over the 70 instances: RE (a) and relative width of the predicted makespan interval (b). Diamonds: medians; labels: Mann–Whitney test between adjacent arms.

The fitness rewards a low worst case *and* a narrow interval, and the weight  $\lambda$  sets the balance between the two. Here  $\lambda = 1$ , and the two bottom rows of Figure 8 show the outcome. Under this objective, the difference in width between the arms with and without the width terminals doubles, from 0.20 points under the makespan objective to 0.42. Rules evolved with the width terminals produce significantly narrower makespan intervals ( $12.04 \pm 0.75$  versus  $12.47 \pm 0.20$  relative width; Mann–Whitney  $z = -2.99$ ,  $p < 0.01$ ,  $|r| = 0.45$ ). In exchange, their expected makespan is higher ( $19.62$  versus  $18.42$  RE;  $z = 3.59$ ,  $p < 0.001$ ,  $|r| = 0.54$ ), which is the trade-off the robust objective is designed to make.

The weight  $\lambda$  of Eq. (13) governs that trade-off, and sweeping it traces the frontier between quality and predictability (Figure 9). Both quantities move monotonically with  $\lambda$ : as the weight grows from 0.5 to 4, the relative width of the makespan interval falls from  $12.18 \pm 0.50$  to  $10.54 \pm 1.26$  while the expected makespan degrades from  $19.20 \pm 1.52$  to  $23.27 \pm 3.80$  RE. At the far end of the sweep the schedules are therefore narrower than any the ablated rules produced, at a cost of about four points of RE.

Sweeping  $\lambda$  for the ablated arm (ten evolutions per value) shows that it has no such curve. Its width stays between 12.43 and 12.56 for every  $\lambda$  from

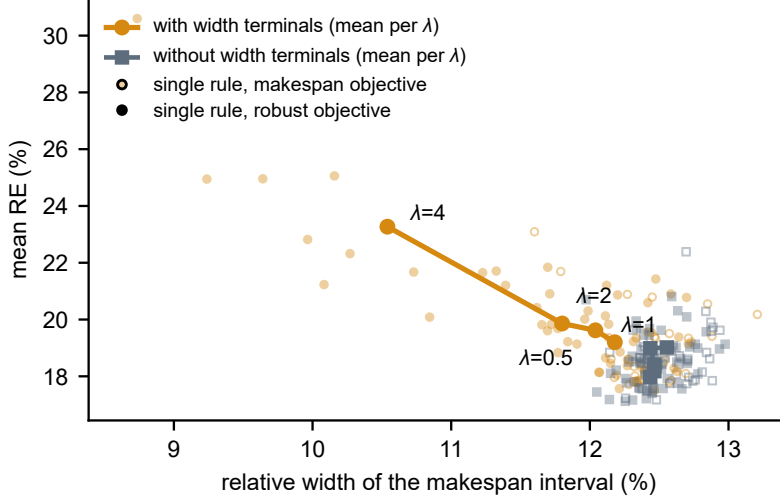


Figure 9: Quality–predictability plane over the 70 instances. Faint markers are evolved rules (amber with width terminals, grey without; open for the makespan objective, filled for the robust one); curves join the arm means along the  $\lambda$  sweep of Eq. (13). The five means without width terminals fall within 0.12 points of width and are unlabeled.

0 to 4, and no individual rule departs from this band, whereas the full arm goes from 12.18 to 10.54. Penalizing width in the fitness therefore yields narrower intervals only when the rule has terminals through which to act on it: the objective works through the representation.

Read at the level of the arm, the price of narrowness rises along the sweep. Between  $\lambda = 2$  and  $\lambda = 4$  the spread between evolutions more than doubles, from 1.74 to 3.80, so the outcome of a single run becomes much less predictable. A weight between 1 and 2 is therefore the default we would recommend. Selecting a rule within an arm amplifies both sides of the trade-off: the narrowest rule of the  $\lambda = 1$  arm on the validation set, used in Section 7.6, reduces  $\bar{\varepsilon}$  by about a fifth at a cost of 7.2 points of RE with respect to the featured rule. For expected makespan alone the smaller terminal set suffices (Section 7.4); the width terminals become necessary when predictability matters.

#### 7.6. Robustness under executional uncertainty

We estimate the robustness measures of Section 5.4,  $\bar{\varepsilon}$ ,  $|\Delta|$  and the  $\text{CVaR}_{0.95}$  of the overrun, by Monte Carlo. We draw  $K=1000$  realizations

Table 8: Predicted quality and executional robustness over the 70 instances: RE, relative width of the predicted interval,  $\bar{\varepsilon}$  ( $\times 10^3$ ) at nominal widths and widened by +20% and +40%, and  $|\Delta|$  at nominal widths. Representatives: the featured rule; for the robust arms, the narrowest rule on the validation set; for the ablated arms, the best rule on the 70 instances, which favors them. Supplementary Section S2 compares all 30 rules of each arm.

| Method                             | RE (%) | Width (%) | $\bar{\varepsilon}$ ( $\times 10^3$ ) |      |      | $ \Delta $ |
|------------------------------------|--------|-----------|---------------------------------------|------|------|------------|
|                                    |        |           | +0%                                   | +20% | +40% |            |
| GP, makespan                       | 17.71  | 12.28     | 5.70                                  | 6.64 | 7.57 | 12.30      |
| GP, makespan, no widths            | 17.17  | 12.48     | 5.75                                  | 6.80 | 7.83 | 12.25      |
| GP, robust $\lambda=1$             | 24.95  | 9.24      | 4.52                                  | 5.32 | 6.08 | 10.31      |
| GP, robust $\lambda=1$ , no widths | 17.45  | 12.05     | 5.54                                  | 6.50 | 7.42 | 12.08      |
| GP, robust $\lambda=4$             | 28.27  | 8.61      | 4.21                                  | 5.01 | 5.73 | 9.78       |
| G&T-MWKR                           | 29.50  | 13.77     | 6.29                                  | 7.51 | 8.58 | 14.94      |
| EST                                | 42.30  | 13.37     | 5.66                                  | 6.79 | 7.83 | 14.64      |

of the durations uniformly on each operation’s interval, execute the fixed processing order on each of them, and use the same realizations for every method. The sampling only evaluates schedules that are already built; the problem, the fitness and the evolution remain free of any distributional assumption. The uniform law is the least committal choice on a bounded support, and three other laws are considered in Supplementary Section S2. To probe behavior under growing uncertainty we also widen every interval symmetrically about its midpoint by +20% and +40% (clipping negative bounds to zero), re-evaluating the same processing orders.

Table 8 reports  $\bar{\varepsilon}$  for a representative rule of each arm and for the two strongest baselines. Under the makespan objective, the featured rule and its ablated counterpart are statistically indistinguishable (5.70 against 5.75 at the nominal width; paired Wilcoxon over the 70 instances,  $z = -0.17$ ). Over all 30 rules of each arm, however, the width terminals reduce the mean  $\bar{\varepsilon}$  from 5.77 to 5.67 (Mann–Whitney  $z = -2.40$ ,  $p < 0.05$  after Holm’s correction; Supplementary Section S2). The rule is more robust than G&T-MWKR (6.29,  $z = -4.42$ ,  $p < 0.001$ ,  $|r| = 0.61$ ) and as robust as EST (5.66,  $z = 0.42$ ), whose predicted makespan is far longer. A low deviation does not require a low makespan, so robustness must be optimized directly, and the robust fitness achieves this. Over all 30 rules of each arm it lowers the mean  $\bar{\varepsilon}$  from 5.67 to 5.50 (Mann–Whitney  $z = -2.68$ ,  $p < 0.05$  after

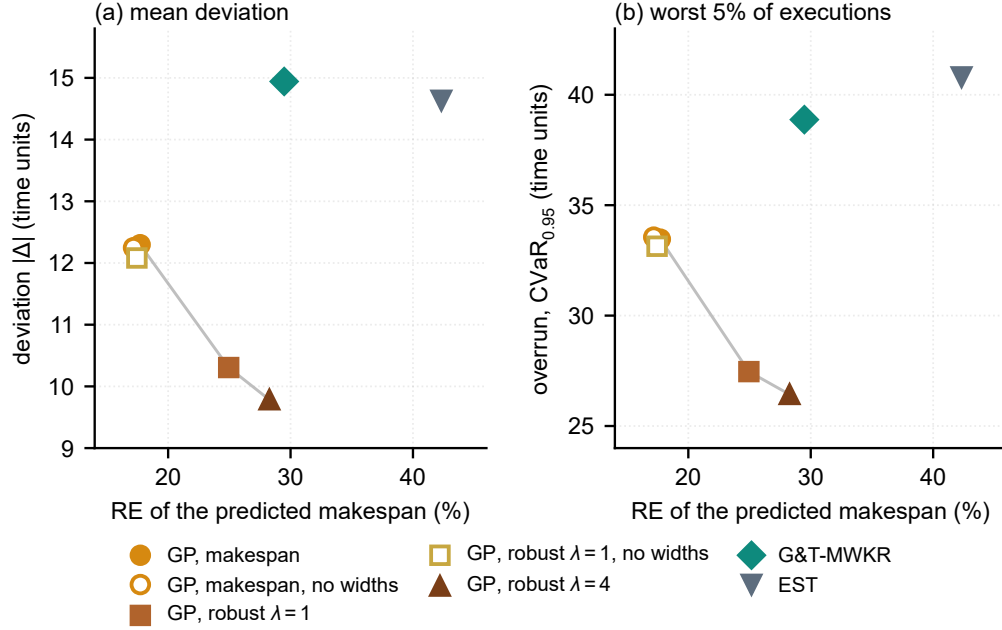


Figure 10: Prediction quality against execution reliability over the 70 instances (uniform law): (a) mean absolute deviation of the executed makespan from the predicted one, (b) mean overrun in the worst 5% of executions, both in time units. The grey line joins the rules with width terminals as  $\lambda$  grows.

Holm’s correction). The narrowest rule of the  $\lambda=1$  arm on the validation set reduces it by about a fifth, from 5.70 to 4.52 ( $z = -6.40$ ,  $|r| = 0.88$  against the makespan rule). The reduction requires the width terminals: under the same fitness, the ablated counterpart reaches 5.54 ( $z = -6.25$ ,  $|r| = 0.86$ ). At  $\lambda=4$ ,  $\bar{\varepsilon}$  falls to 4.21. The ordering holds as the intervals are widened, and the advantage of the robust rule grows with the widening: at +40% its  $\bar{\varepsilon}$  is 6.08, against 7.57 for the makespan rule and 8.58 for G&T-MWKR.

Figure 10 complements  $\bar{\varepsilon}$  with the deviation in time units,  $|\Delta|$ , against the quality of the prediction. The rule deviates by 12.30 time units, against 14.64 for EST and 14.94 for G&T-MWKR. The robust rules move down and to the right along the grey line: the narrowest rule of the  $\lambda=1$  arm deviates by 10.31 time units, 16.2% less than the makespan rule ( $z = -5.55$ ,  $p < 0.001$ ,  $|r| = 0.76$ ), and the rule at  $\lambda=4$  by 9.78. At matched a-priori quality the gain is clearest: the  $\lambda=4$  rule and G&T-MWKR predict almost

the same makespan, 28.27 against 29.50 points of RE, and the rule deviates 34.5% less, on 69 of the 70 instances ( $z = -7.26$ ,  $p < 0.001$ ,  $|r| = 1.00$ ). Over the worst 5% of executions (Figure 10b) the ordering is the same, and the robust objective reduces the overrun beyond the prediction in about the same proportion as the mean deviation, from 33.46 to 27.46 time units at  $\lambda=1$ . In absolute terms, the length of those executions follows the expected makespan that the robust rules trade for reliability.

Supplementary Section S2 reports the per-instance distributions, the full comparison over all 30 rules of each arm, and three other realization laws, under which the ordering and every significant contrast above hold.

### 7.7. *Asymmetric intervals*

All the intervals used so far are symmetric about the crisp duration. To test whether the contribution of the width terminals depends on that construction, the four arms of Sections 7.4 and 7.5, both objectives with and without width terminals, are re-evolved with fifteen seeds on the right-skewed versions of TA11–TA14 (Section 5.1) and evaluated on the 70 right-skewed instances (Supplementary Section S3). The outcome matches the symmetric case. Under the robust objective the width terminals narrow the predicted interval from 12.97% to 12.67% of the expected makespan and reduce the deviation of the executed makespan,  $|\Delta|$ , from 13.03 to 12.75 time units, both significant after Holm’s correction; under the makespan objective neither difference is significant. The robustness that the width terminals provide is therefore not a product of symmetric intervals.

### 7.8. *Limitations*

Four limitations bound the scope of these conclusions. First, the rules are evolved and applied with the semi-active decoder. Run inside Giffler and Thompson’s active decoder after evolution, the same rules reach 23.18% RE instead of 18.99% (Section 7.3); rules evolved directly inside the active decoder have not been tested.

Second, the rules are evolved under the symmetric  $\pm 15\%$  interval scheme standard in the IJSP literature. Their robustness ordering holds under intervals widened by 20% and 40% (Section 7.6), and re-evolution on right-skewed intervals preserves the contribution of the width terminals (Section 7.7), but evolution under intervals of a different magnitude has not been tested.

Third, budgets are measured in seconds of a Python implementation shared by all methods. In a compiled language such as C++ the curves of

Section 6.5 would converge at lower budgets; the comparison itself changes only if the language alters the cost of a schedule built by the rule relative to one decoded by the genetic algorithm. The rankings of the other sections do not depend on time.

Finally, no learned policy is among the baselines. The only reinforcement-learning method published for the IJSP [39] leaves part of the sequencing to be resolved during execution, so it does not belong to the single-pass class compared here, and it reports results on a different selection of classical instances.

## 8. Conclusions and Future Work

We presented a GP hyper-heuristic for job shop scheduling with interval durations, which evolves priority rules over a terminal set exposing both the worst-case values and the widths of the uncertain quantities. The evolved rules are compact enough to be read and simplified by hand. In a single constructive pass they average 18.99% RE over 30 evolutions, and the featured rule attains 17.71%, against 29.5% for the strongest baseline, G&T-MWKR; the featured rule outperforms every deterministic constructive baseline on all 70 interval Taillard instances. Evolved on four  $20 \times 15$  instances, the rules transfer without retraining to every size class and to the classical instances, and perform best on the unseen  $50 \times 15$  class.

The questions of Section 1 receive the following answers. First, the evolution selects the classical scheduling attributes most often, slack, work remaining, earliest start and processing time, and keeps the width terminals in most rules. Second, under the default expected-makespan objective the magnitude of the uncertainty is not needed to dispatch well: rules evolved without the width terminals, or on the crisp midpoint instances, perform as well as the interval-aware ones. Third, an objective that rewards a narrow makespan interval leads to more predictable executions. With  $\lambda = 1$  the predicted intervals become narrower and the executions deviate less from them: over the 30 rules of each arm the mean relative deviation  $\bar{\epsilon}$  falls from 5.67 to 5.50, and the narrowest rule on the validation set reduces it by about a fifth, a reduction that extends to the worst 5% of executions and grows as the intervals are widened. The expected makespan rises in exchange, by an amount that the weight  $\lambda$  controls. The reduction requires the width terminals, on symmetric and on asymmetric intervals alike, so it is under this objective that the magnitude of the uncertainty becomes useful information.

Fourth, the place of evolved rules among per-instance methods depends on the budget. Compared with our implementation of the genetic algorithm published for the IJSP, which shares decoder, evaluator and language with the rule, GP-best-of- $N$  attains the lowest RE up to a budget that grows with instance size, from about 10 s per instance on the classical instances to about 300 s on  $50 \times 15$ , and a single pass delivers in milliseconds a quality that the genetic algorithm needs between 2 and 46 s per instance to reach. As a seed, the rule speeds up the early search of the genetic algorithm. At larger budgets the genetic algorithm attains a lower RE, and the published ABC variants, which combine population search with local search, obtain the best results. Evolved rules are therefore a high-quality, interpretable method that decides in milliseconds and complements per-instance search, suited to settings in which schedules must be built or rebuilt quickly.

Future work follows four lines. First, refining the numeric weights of evolved rules, which the sensitivity analysis of Section 7.1 shows the evolution does not always leave at their best values; since only the relative weights of the terms affect the dispatching decisions, a local search over them is the natural complement. Second, evolving rules inside Giffler and Thompson’s active decoder and under intervals of other magnitudes, the two settings that Section 7.8 leaves open. Third, seeding the published ABC variants with the rule and with the diverse pools of sequences that GP-best-of- $N$  produces, and richer symbolic models such as ensembles of complementary rules [32]. Fourth, a controlled comparison with learned neural constructive policies under matched training and inference budgets, the evaluation that the recent literature [36] identifies as missing.

### **CRedit authorship contribution statement**

**Hernán Díaz:** Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Resources, Data curation, Writing – original draft, Writing – review & editing, Visualization.

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## **Declaration of competing interest**

The author declares that there are no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## **Data availability**

The GP hyper-heuristic is implemented from scratch in pure Python/NumPy, with no evolutionary-computation dependencies; evolved rules are stored as plain JSON expression trees and evaluated as drop-in dispatching strategies. All benchmark instances, the 430 evolved rules of every experimental arm, the primary result files behind each table and figure, and a self-contained implementation that re-derives the deposited results from scratch are openly available in a Zenodo repository [51] (doi:10.5281/zenodo.21716972). Version 2.0 of the repository (doi:10.5281/zenodo.23156759) holds the material of this article, including its supplementary material.

## **Declaration of generative AI and AI-assisted technologies in the manuscript preparation process**

During the preparation of this work the author used Claude (Anthropic) in order to assist with the drafting and editing of the manuscript. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the published article.

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